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Expression level dependent changes in the properties of P2X₂ receptors

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Abstract

The currents of P2X₂ receptors expressed in *Xenopus* oocytes or HEK293 cells show significant cell-to-cell variation in many properties including the rate of desensitization and the magnitude of potentiation by zinc or acidic pH. In this study, we examined whether differences in expression levels underlie this variability. We injected *Xenopus* oocytes with different concentrations of RNA encoding rat P2X₂ to give a wide range of maximum current amplitudes, and then measured the potentiation of responses to 10 μ M adenosine 5'-triphosphate (ATP) by zinc or acidic pH.

Individual oocytes showed potentiation ratios that ranged from 1.4- to 25-fold. Oocytes with small amplitude responses to a saturating concentration of ATP tended to have larger potentiation ratios than oocytes with large amplitude responses. This phenomenon was explained by an inverse correlation between the EC₅₀ for ATP and the maximum current amplitude, with the EC₅₀ decreasing from about 37 to 7 μ M as expression level increased. In contrast, the Hill coefficient was not correlated with the maximum current amplitude. Truncated receptors lacking the last 76 amino acids also showed an inverse correlation between the EC₅₀ and the maximum current amplitude. Thus, the interactions that cause expression-dependent changes in P2X₂ receptor properties must involve domains proximal to position H397.

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1. Introduction

P2X receptors are a family of adenosine 5'-triphosphate (ATP)-gated cation channels expressed by neurons and by non-neuronal cells such as muscle cells and epithelial cells (Ralevic and Burnstock, 1998; Khakh, 2001; Robertson et al., 2001; North, 2002). Seven P2X receptor genes (P2X_{1–7}) have been cloned from rat and each gene encodes a single subunit. The assembled receptors appear to be composed of three subunits and each subunit has two transmembrane domains with intracellular N- and C-termini and a single extracellular domain (Nicke et al., 1998; Stoop et al., 1999).

A number of properties of P2X₂ receptors show far more cell-to-cell variability than expected by measure-

ment error alone. Among the properties reported to vary significantly are the extent of inward rectification (Evans et al., 1996; Zhou and Hume, 1998), the rate of desensitization (Zhou et al., 1998), the ability to enter the high conductance 'megapore' state (Virginio et al., 1999) and the extent of potentiation in response to extracellular zinc or acidic pH (Clyne et al., 2002).

The gating kinetics of several potassium channels including *Shaker* H4 (Moran et al., 1992), human HLC3, (Honore et al., 1992), and Kv 1.2 (Guillemare et al., 1992) and *Shaker* B (Chen et al., 2000) have been shown to vary with expression levels in *Xenopus* oocytes. In addition, *Xenopus* oocytes expressing high levels of glycine receptors have smaller EC₅₀ values for glycine than oocytes expressing low levels of glycine receptors (Taleb and Betz, 1994). We therefore decided to test whether differences in expression level could account for the variation in zinc and pH modulation of P2X₂ receptors.

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2. Materials and methods

2.1. Mutagenesis

Rat P2X₂ cDNA in pcDNA1 was obtained from Dr D. Julius, University of California, San Francisco, CA, USA (Brake et al., 1994). A C-terminal deletion mutant P2X₂ receptor was made by using the quickchange site-directed mutagenesis kit (Stratagene, LaJolla, CA, USA) to substitute a stop codon (TAA) for the bases encoding H397. This mutation is referred to as H397#. Mutagenic oligonucleotides were obtained from Operon Technologies, Inc. (Alameda, CA, USA), and contained the base changes necessary for the introduction of a restriction enzyme recognition site. Putative mutants were first screened by restriction enzyme digestion and then confirmed by DNA sequencing (University of Michigan DNA Sequencing Core, Ann Arbor, MI, USA). We also performed a few experiments on the effect of expression level on pH potentiation of point mutants that are unaltered in their median response to pH (H174A, H213A, H245A) or on zinc potentiation of point mutations that are unaltered in their median response to zinc (H174A, H245A, H319A). The properties of these mutations are characterized in greater detail in Clyne et al. (2002).

2.2. Expression of P2X₂ receptors

P2X₂ receptors were expressed in defolliculated stage V–VI *Xenopus laevis* oocytes. Oocytes were harvested by procedures approved by the University of Michigan Committee on the Use and Care of Vertebrate Animals

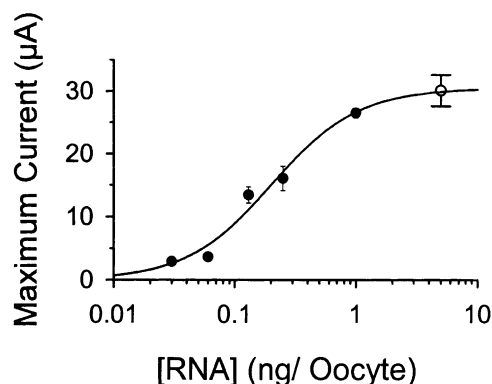


Fig. 1. Relation between maximum responses and amount of RNA injected. The current amplitude at -50 mV in response to a saturating concentration of ATP ($200 \mu\text{M}$) increased with increasing amounts of P2X₂ RNA injected per oocyte, but eventually saturated. All currents were inward, but for convenience the maximum current is expressed as a positive number. The solid circles represent data obtained 2 days after injection of the indicated amounts of RNA into oocytes from a single frog. The open circle indicates the average response of oocytes from another frog also tested at 2 days after RNA injection. The solid line is the fit of the data to the logistic equation. At least five oocytes were tested at each concentration and for some points the error bars (SEM) were smaller than the symbol.

and have been described in detail (Zhou and Hume, 1998). Frogs were anesthetized by immersion in well water containing 3-aminobenzoic acid ethyl ester (1 g l^{-1} , Sigma) and were humanely killed after the final collection of oocytes.

P2X₂ and mutant receptor RNA were synthesized using the T7 mRNA message machine kit from Ambion (Austin, TX, USA) and were quantitated both spectroscopically and on ethidium bromide gels. We selected the amount of RNA to inject based on the data illustrated in Fig. 1. These data show that at a holding potential of -50 mV, the maximum current amplitude at 48 h after injection saturated at approximately $-29 \mu\text{A}$ for oocytes injected with more than 1 ng of RNA. Half-saturation occurred at approximately 0.2 ng per oocyte. Following injection with RNA, oocytes were kept at 17°C until the initial recording session, typically 40–48 h after injection. Beginning approximately 48 h after injection, oocytes were stored in a refrigerator at 6°C and additional recordings were made up to 5 days after injection. When treated in this manner, the maximum expression level did not change significantly when studied at different times after injection.

For each experiment, oocytes were injected with 3–5 different concentrations of RNA ranging from 20 pg to 2 ng of RNA per oocyte. These concentrations were selected to yield a wide range of expression levels and responses were characterized electrophysiologically 2–5 days after injection.

2.3. Solutions

Our oocyte recording solution (ORS) contained (in mM): 90 NaCl , 1 KCl , 1.3 MgCl_2 , 10 HEPES pH 7.5. At high ATP concentrations, we compensated for the chelation of Mg^{2+} by ATP by adding MgCl_2 to our solutions such that all solutions contained 1 mM free Mg^{2+} , as determined by the program bound and determined (Brooks and Storey, 1992). Our ATP stock solution consisted of disodium ATP (Sigma-Aldrich, St Louis, MO, USA) dissolved at 100 mM in double-distilled H_2O , aliquoted and stored at -20°C for 1–3 months. The ATP concentrations of all solutions were verified by taking spectroscopic measurements at 259 nm . A 1 M stock of ZnCl_2 was dissolved in acidic, double-distilled H_2O in order to prevent precipitation. For recording, ATP stock solutions were diluted in ORS with the required zinc and magnesium then the pH was adjusted to the required level. All solutions were used within 48 h.

2.4. Electrophysiological recordings and data analysis

P2X₂ and mutant receptor currents were investigated by using two-electrode voltage clamp with an np1 TurboTec 3. All experiments were done at room temperature (22°C) and the holding potential was -50 mV . All cur-

rents were inward, but for convenience when we plotted the maximum current, it is shown as a positive quantity (i.e. the absolute value of the actual peak current).

Averaged data were expressed as the mean $\pm$ SEM (standard error of the mean). The program SIGMAPLOT 7.0 was used for curve fitting and statistical analysis. Concentration–response relations were fit using the logistic equation,

$$I = \frac{I_{\max}}{1 + \left(\frac{EC_{50}}{[ATP]}\right)^H}$$

where H is the Hill coefficient and EC_{50} is the concentration that elicited a half-maximal response. For the

experiments shown in Fig. 3(F) and (G) the EC_{10} concentration of ATP was calculated with the equation $EC_{10} = EC_{50}(10/90)^{1/H}$. The range of EC_{10} concentrations used in these experiments was 3–13 μ M in 1 μ M increments. Thus, the calculated EC_{10} was rounded to the nearest micromolar concentration. As noted in Section 3, this rounding is expected to introduce some variation into the measured potentiation ratios.

Data were fit using the curve-fitting module of SIGMAPLOT 7.0. The criterion for accepting a fit was $p < 0.01$ that the fitted function and the actual values of the dependent variable corresponded by chance alone. For simplicity, data were initially fit with a linear regression. The scatter plots of potentiation ratios with 10 μ M ATP (Figs. 2(E), 3(D) and 4(C)) and EC_{50} values (Figs. 3(B) and 4(A)) were also fit with the following functions derived from the two models described in Section 4. Model 1 predicts that the relation between the EC_{50} (y) and expression level (x) will be constant at the maximal EC_{50} until a binding protein is saturated and then will fall along a hyperbolic trajectory according to the equation:

$$y = \text{Min} + \left[\frac{B}{B + x} \right] (\text{Min} - \text{Max})$$

where Min is the EC_{50} for unbound receptors, Max the EC_{50} for bound receptors, and B is a constant related to the concentration of the binding protein and the strength of the binding reaction. Model 2 predicts that as receptors dimerize, the EC_{50} will fall along a trajectory

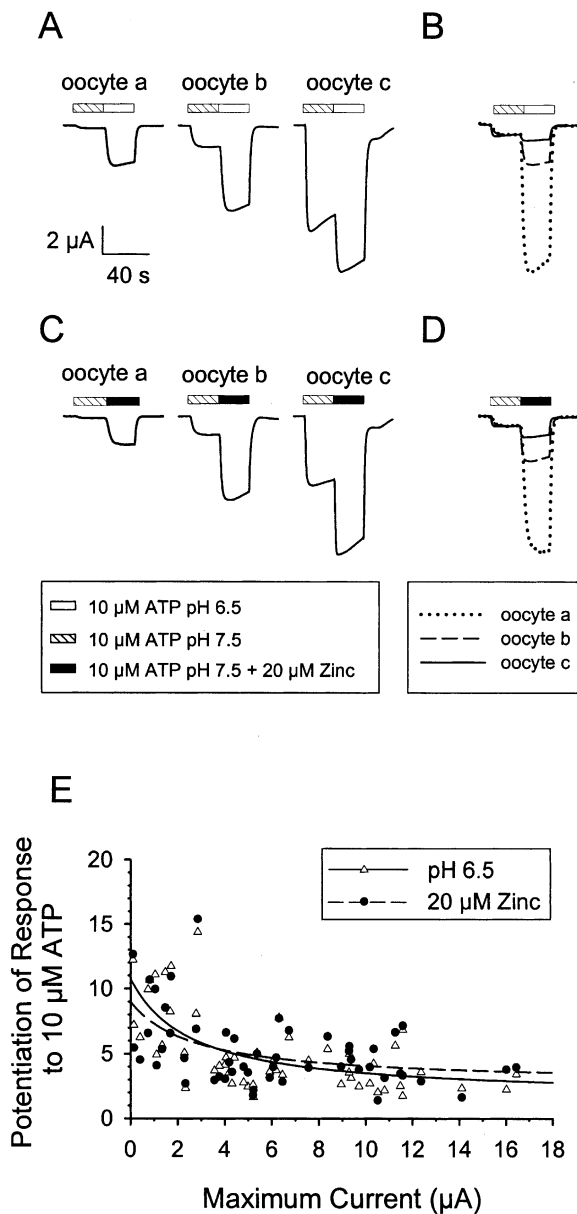


Fig. 2. Inverse correlation of zinc and pH potentiation of P2X₂ receptors with maximum current amplitude. All data illustrated are from oocytes from a single frog studied at 2 days after RNA injection. High, medium, and low levels of maximal current were obtained by injecting three different concentrations of RNA. (A) The responses of three oocytes (a, b, c) expressing P2X₂ receptors to 10 μ M ATP at pH 7.5 followed by 10 μ M ATP at pH 6.5. The calibration bar below A applies to A and C and the legend below C applies to A–D. (B) The responses shown in A normalized so that the initial response to 10 μ M ATP was the same amplitude in all three traces. The legend below D applies to both B and D. (C) The responses of the same three oocytes shown in A to 10 μ M ATP at pH 7.5 followed by 20 μ M zinc plus 10 μ M ATP at pH 7.5. (D) The responses shown in C normalized so that the initial response to 10 μ M ATP was the same amplitude in all three traces. (E) The zinc (filled circles) or pH (open triangles) potentiation ratios for single oocytes were plotted against the current amplitude in response to a saturating concentration of ATP (200 μ M). The zinc potentiation ratio is the ratio of the current amplitude in response to 10 μ M ATP plus 20 μ M zinc to the response to 10 μ M ATP alone. The pH potentiation ratio is the ratio of the response of the receptors to 10 μ M ATP at pH 6.5 to the response to 10 μ M ATP at pH 7.5. The lines shown represent fits with a hyperbolic decay function. For pH, the maximum ratio was 10.8, the minimum ratio was 1.6 and the constant that determines the rate of decay (B) was 2566 nA ($R^2 = 0.44$). For zinc, the maximum ratio was 9.0, the minimum ratio was 2.7 and the B constant was 2711 nA ($R^2 = 0.24$). Both zinc and pH potentiation ratios also showed a significant linear correlation with maximum currents (for pH potentiation, slope = $-0.41 \mu\text{A}^{-1}$, intercept = 7.7, $R^2 = 0.33$; for zinc potentiation, slope = $-0.27 \mu\text{A}^{-1}$, intercept = 6.9, $R^2 = 0.18$).

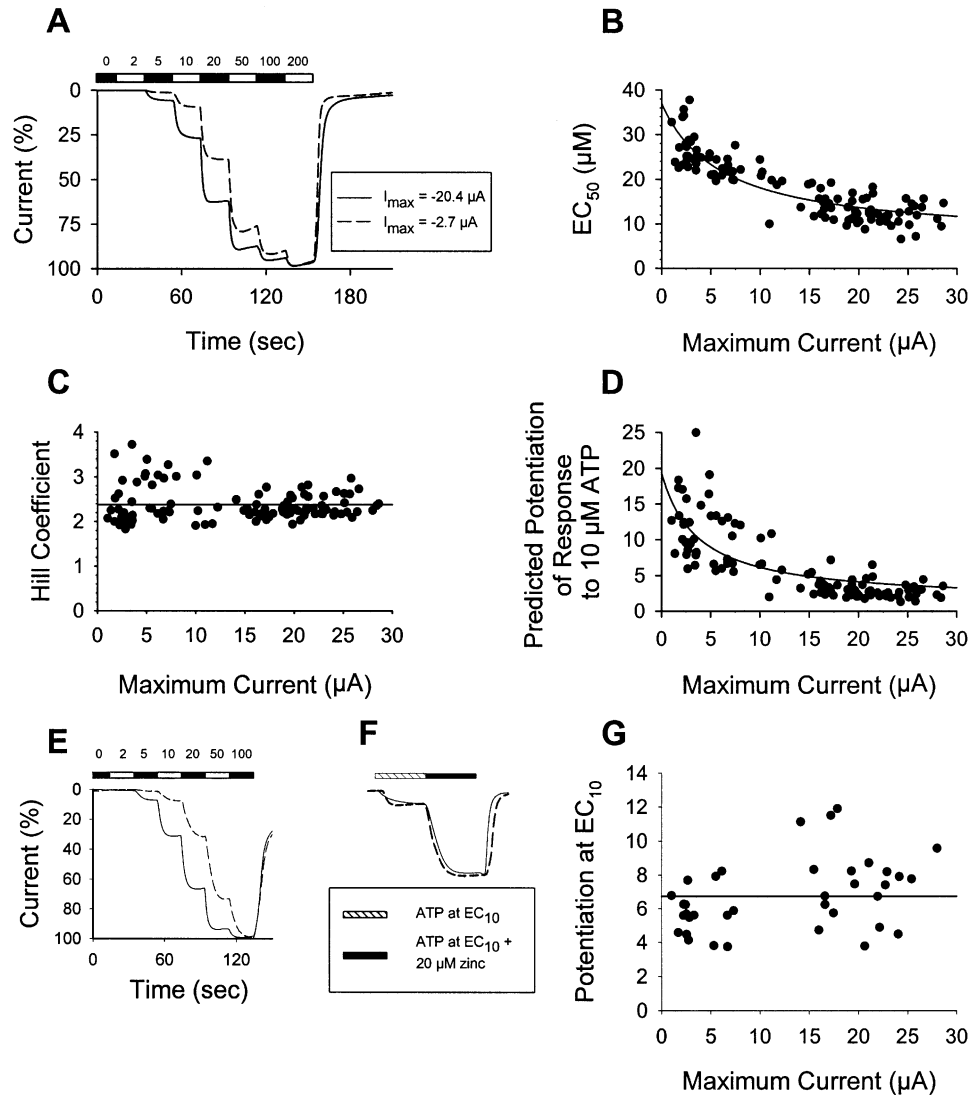


Fig. 3. The EC_{50} of the ATP concentration–response relation was inversely proportional to the maximum current amplitude. The concentration–response relations of $P2X_2$ receptors were determined for oocytes injected with different amounts of RNA. Data shown in B, C and D were obtained in four recording sessions, all of which showed similar trends. (A) The current response of two oocytes expressing $P2X_2$ receptors to sequential application of increasing concentrations of ATP (indicated by the black and white bars). The current responses were normalized to the largest current for each oocyte. The concentration–response relation of the oocyte illustrated with the solid line had a maximum current of $-20.4 \mu A$, an EC_{50} of $16 \mu M$ and a Hill coefficient of 2.2. This concentration–response relation was left-shifted compared to the concentration–response relation of the other oocyte (dashed line), which had a maximum current of $-2.7 \mu A$, an EC_{50} of $25 \mu M$, and a Hill coefficient of 2.2. (B) The EC_{50} of the ATP concentration–response relation was inversely proportional to the maximum current amplitude of individual oocytes. The solid curve corresponds to a fit to a hyperbolic decay from an EC_{50} of 37.0 – $6.6 \mu M$ with a B constant equal to 6108 nA ($R^2 = 0.78$). The best linear fit to these data had a maximum EC_{50} of $28.2 \mu M$ and a slope of $-0.7 \mu M \mu A^{-1}$ ($R^2 = 0.74$). Both fits indicated a significant inverse correlation. (C) The Hill coefficient of the ATP concentration–response relation was independent of the maximum current amplitude. Linear regression showed no trend in the data, therefore the line is drawn at the average (2.4) for all of the data. (D) The predicted potentiation ratio at $10 \mu M$ ATP for each oocyte studied in B and C. These ratios were calculated by dividing 100% (the maximum response) by the predicted response to $10 \mu M$ ATP at pH 7.5 based upon the parameters of the concentration–response relations shown in panel B and C. A linear regression fit to these data gave a maximum predicted potentiation of 12.4 and a slope of $-0.4 \mu A^{-1}$ ($R^2 = 0.61$) and a hyperbolic decay (the curve illustrated) gave a maximum potentiation of 19.3, a minimum potentiation of 1.3 and a B constant equal to 3664 nA ($R^2 = 0.60$). (E) The current responses of two oocytes expressing $P2X_2$ receptors to sequential application of increasing concentrations of ATP (indicated by the black and white bars) normalized to the maximum current recorded from each oocyte. The maximum response for the cell depicted by the solid line was $-17 \mu A$ and the maximum response for the cell depicted by the dashed line was $-2 \mu A$. (F) Potentiation of the responses of the two oocytes illustrated in E by $20 \mu M$ zinc when ATP was applied at the EC_{10} estimated from fits to the concentration–response data for the two oocytes depicted in E. The EC_{10} for the cell depicted by the solid line was $6 \mu M$ and the EC_{10} for the cell depicted by the dashed line was $11 \mu M$. (G) Potentiation ratios for a series of cells tested at the EC_{10} . The experimental protocol was as illustrated in E and F. Linear regression showed no trend in the data, therefore the line is drawn at the average (6.7) for all of the data.

described by the root of a quadratic according to the equation:

$$y = \text{Min} + \left[\frac{-K + \sqrt{K^2 + 4xK}}{2x} \right] (\text{Max} - \text{Min})$$

where Min represents the EC_{50} of the receptors that have dimerized, Max represents the EC_{50} of receptors that have not dimerized, and K is constant related to the affinity of the receptors for each other and the surface area of the oocyte. Both B and K determine how rapidly the respective functions decline, but explicitly relating these constants to the properties of the proposed binding reactions would require several additional assumptions, and so was not attempted. With B or K free to vary and Min constrained to be no smaller than the lowest measured y value, either function fit the data equally well and

gave similar values for Max and Min, so only the fits to the hyperbolic functions are shown. The parameters of the hyperbolic and linear regressions are given in the figure legends, and the quality of these fits is indicated by the R^2 values. For the plots of Hill coefficients (Figs. 3(C) and 4(B)) and potentiation ratios at the EC_{10} (Fig. 3(G)), all regressions tested had $p > 0.01$, indicating no correlation between EC_{50} values or potentiation ratios and expression levels. We therefore displayed a straight line at the average value in these plots.

3. Results

3.1. Zinc and pH potentiation ratios were inversely correlated with current amplitude

To determine whether the variation in zinc and pH modulation of P2X_2 receptors depended on the level of expression, we injected oocytes with several different concentrations of RNA encoding rat P2X_2 receptors. Fig. 2(A) shows the responses of three oocytes (oocyte *a*, oocyte *b*, and oocyte *c*) expressing P2X_2 receptors at different levels to 10 μM ATP at both pH 7.5 and pH 6.5. Decreasing the pH to 6.5 potentiated the responses of all three oocytes, but the potentiation ratio (peak current at pH 6.5 divided by peak current at pH 7.5) was very different for each oocyte (oocyte *a* = 14.3, oocyte *b* = 4.0, oocyte *c* = 1.4). The wide range of potentiation is more readily visualized when the initial response to 10 μM ATP is normalized to 100% (Fig. 2(B)). Because

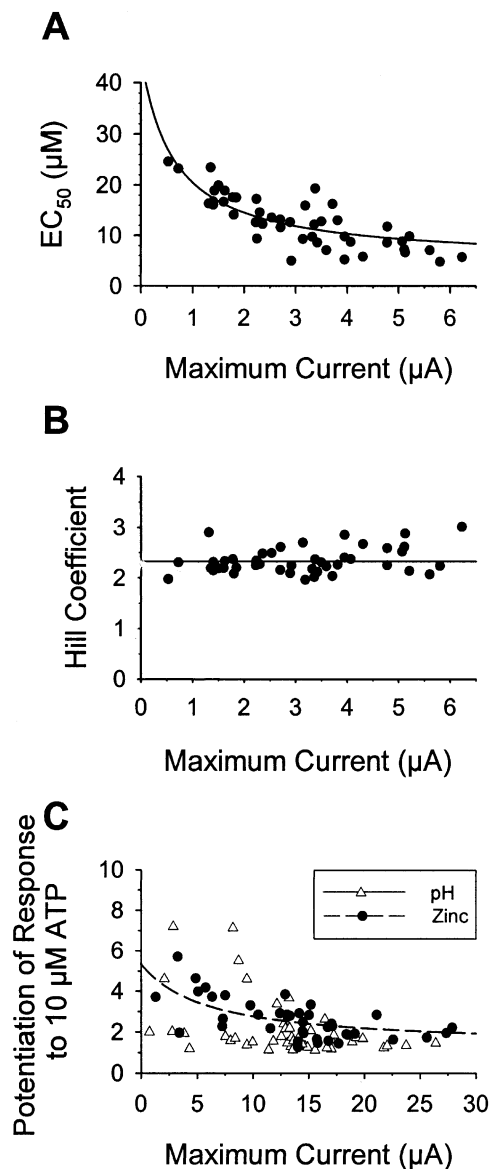


Fig. 4. The C-terminal 76 amino acids were not required for expression-dependent changes in P2X_2 receptor properties. Oocytes were injected with different amounts of RNA encoding H397#, a truncated form of the P2X_2 receptor. (A) The EC_{50} of the ATP concentration–response relation was inversely proportional to the maximum current amplitude of individual oocytes. These oocytes were studied 2 days after RNA injection and came from a single frog. The solid curve corresponds to the hyperbolic decay function (Maximum = 45.1 μM , Minimum = 4.8 μM , $B = 637\text{nA}$, $R^2 = 0.66$). Linear regression also indicated a significant inverse correlation with a slope of $-2.9 \mu\text{M} \mu\text{A}^{-1}$ and intercept of 21.7 μA ($R^2 = 0.64$). (B) The Hill coefficient of the ATP concentration–response relation was independent of the maximum current amplitude. These data are for the same H397# oocytes as shown in A. Linear regression showed no trend in the data, so the line is drawn at the average (2.3) for all of the data. (C) Zinc (filled circles) or pH (open triangles) potentiation ratios for single oocytes plotted against the maximum current amplitude for the H397# C-terminal deletion mutant. The data illustrated came from experiments on oocytes from several different frogs studied at 2–4 days after RNA injection. The line through the zinc data represents the fit to the hyperbolic decay function (maximum ratio = 5.3, minimum ratio = 1.4, $B = 5809\text{nA}$, $R^2 = 0.42$). No line is shown for the pH data because the best hyperbolic fit (maximum ratio = 4.2, minimum ratio = 1.1, $B = 5618\text{nA}$, $R^2 = 0.15$) was only significant at the 5% level. Linear regression demonstrated a significant inverse correlation with maximum currents (for pH potentiation, slope = $-0.11 \mu\text{A}^{-1}$, intercept = 3.5, $R^2 = 0.18$; for zinc potentiation, slope = $-0.1 \mu\text{A}^{-1}$, intercept = 4.0, $R^2 = 0.41$).

these oocytes were from the same frog and recorded on the same day, this variation could not be due to variations among different batches of oocytes or to variation between the test solutions. Another modulator, 20 μ M zinc, induced a similar wide range of potentiation of the responses to 10 μ M ATP at pH 7.5 (Fig. 2(C)–(D)) in the same three oocytes (oocyte *a* = 13.7, oocyte *b* = 4.5, oocyte *c* = 2.0). The amplitudes of the responses to a saturating amount of ATP at pH 7.5 (200 μ M) were -2.9 μ A (oocyte *a*), -6.1 μ A (oocyte *b*), and -10.5 μ A (oocyte *c*). Thus, the oocyte with the largest response to 200 μ M ATP and presumably the most receptors (oocyte *c*) had the smallest potentiation ratio. Conversely, the oocyte with the smallest response to 200 μ M ATP and the fewest receptors (oocyte *a*) had the largest potentiation ratio.

The relation between maximum current amplitude and potentiation ratios for a number of oocytes expressing P2X₂ receptors (taken from a single frog and studied on the same recording day) is shown in Fig. 2(E). Although the potentiation ratios varied quite a bit for oocytes with similar maximum currents, oocytes with small responses to 200 μ M ATP tended to have larger potentiation ratios than oocytes with large responses to 200 μ M ATP. Indeed, a linear regression fit to the zinc and pH data showed a significant inverse correlation. However, a linear model cannot represent the true form of the data, since the potentiation is never expected to fall below 1. We also fit the data with a hyperbolic decay function (curves in Fig. 2E) and with a function that is the root to a quadratic equation (see Section 2) because the models considered in Section 4 predict the data might follow these functions. These fits also gave support for a significant inverse correlation between potentiation ratios and maximum currents. Similar results (potentiation ratios to 10 μ M ATP varying from 1.3- to 25-fold and inversely correlated with expression level) were obtained from all seven experiments in which the pH or zinc potentiation of wild type P2X₂ receptors was studied (although no other experiment involved as many oocytes from a single day as the one illustrated in Fig. 2). We also examined the effects of expression level on pH and zinc potentiation of a few point mutants that were unaltered in their median response to pH (H174A, H213A, H245A) or zinc (H174A, H245A, H319A) (Clyne et al., 2002). Like wild type receptors, the potentiation ratios of these mutants also ranged from less than 2-fold to more than 20-fold and were inversely correlated with maximum currents.

3.2. The EC_{50} and maximum response of P2X₂ receptor are inversely related

Potentiation of P2X₂ responses by zinc and acidic pH is manifest as a shift in the EC_{50} of the concentration–response relation, with little or no change in the

maximum current (King et al., 1996, 1997; Nakazawa et al., 1997; Stoop et al., 1997; Clyne et al., 2002). The amount of potentiation that one can observe therefore depends critically on the concentration of ATP used. For instance, if one uses the EC_{50} concentration of ATP, then the maximum potentiation attainable is $100/50 = 2$ -fold. In contrast, if one uses the EC_5 concentration of ATP, up to 20-fold potentiation can be attained. Thus, one possible explanation for why the potentiation ratios varied could be that the concentration–response relations for P2X₂ receptors varied among oocytes.

To test this possibility, we injected oocytes with different amounts of RNA to give a wide range of maximum currents and then sequentially applied a series of increasing concentrations of ATP (2–200 μ M) to determine the concentration–response relation for each oocyte. Fig. 3(A) shows the responses of two oocytes expressing different amounts of P2X₂ receptors to sequential applications of increasing concentration of ATP. The oocyte that had the larger maximum response to ATP had a smaller EC_{50} compared to the other oocyte. The relation between the maximum current amplitude and EC_{50} is summarized for a number of oocytes in Fig. 3(B). The linear, hyperbolic and quadratic fits all indicated a significant inverse correlation. In contrast, the Hill coefficient was not affected by changes in receptor expression levels (Fig. 3(C)), nor was there any apparent correlation between the rate of desensitization to 200 μ M ATP and the maximum currents (data not shown).

We calculated the predicted potentiation ratio by dividing 100% (the maximum response) by the predicted percent response to 10 μ M ATP based upon the EC_{50} and Hill coefficient for each oocyte. For example, if 10 μ M ATP corresponded to the EC_8 of a particular cell, the predicted potentiation ratio for that cell was $100/8 = 12.5$. Fig. 3(D) shows that the range of predicted potentiation ratios extended over nearly the same range as the experimental data on potentiation shown in Fig. 2(E), and showed a similar inverse correlation with the level of expression. This suggested that the observed change in EC_{50} could fully account for the large range of potentiation ratios observed.

To test this idea, we measured zinc potentiation in individual oocytes studied as close as possible to their actual EC_{10} , rather than with 10 μ M ATP. If changes in EC_{50} with expression level fully account for the changes in potentiation, then potentiation should not correlate with maximum currents in this experiment. For each oocyte, we first determined the concentration–response relation for ATP in the absence of zinc, and then tested potentiation by 20 μ M zinc at the concentration of ATP (rounded to the nearest 1 μ M) that was determined to be the EC_{10} for that oocyte (Fig. 3(E) and (F)). As predicted, when oocytes were studied close to their actual EC_{10} there was no longer any significant correlation between the potentiation observed and the maximum

current amplitudes (Fig. 3(G)), but considerable variability remained. In an attempt to understand this residual variation, we did an additional series of experiments. For a group of seven oocytes in which the estimated EC_{10} was 4 or 5 μ M, we tested the potentiation by 20 μ M zinc observed for 3, 4, 5 and 6 μ M ATP. The average potentiation to 3 μ M ATP was 13.6 ± 1.1 and the average potentiation for 6 μ M ATP was 5.1 ± 0.4 . Simple calculations show that for a cell with a true EC_{10} of 4.5 μ M and a Hill coefficient of 2.5, using 4 μ M ATP would predict 13-fold potentiation and using 5 μ M ATP would predict 8-fold potentiation. Thus residual variation in the measured potentiation ratios was unavoidable due to small errors in estimating the EC_{10} by least squares fitting and rounding off the concentration to the nearest micromolar in order to make the experiment manageable.

3.3. The C-terminus is not required for expression-dependent effects

Increasing receptor expression levels might lead to changes in the response of $P2X_2$ receptors due to saturation of an endogenous binding protein present at limited concentration or due to receptor–receptor interactions (see Section 4). A number of ionotropic glutamate receptors and potassium channels interact with intracellular scaffolding proteins at their extreme C-terminal end (Sheng and Sala, 2001; O'Brien, 1998). Thus, to begin investigating the molecular basis for these expression-dependent effects on $P2X_2$ receptors, we created a C-terminal deletion mutant (H397#) by substituting a stop codon for the bases encoding histidine 397. This deletion mutant removed the last 76 amino acids of the carboxyl terminus, thereby eliminating a phosphorylation site for cAMP-dependent protein kinase (S431) (Chow and Wang, 1998), a putative protein kinase C phosphorylation site (S426), three putative casein kinase II phosphorylation sites (T413, S457, and T462), and a proline-rich region (403–427).

We injected oocytes with five different concentrations of mutant RNA to give a wide range of maximum currents. After two days of incubation, we measured the response of individual oocytes to a series of increasing concentrations of ATP (1–100 μ M). Like oocytes expressing wild type $P2X_2$ receptors, the EC_{50} values for oocytes expressing the H397# truncation varied widely (from 5 to 25 μ M) and were inversely related to the level of expression (Fig. 4(A)). Also similar to oocytes expressing wild type $P2X_2$ receptors, the Hill coefficients for oocytes expressing H397# were independent of expression level (Fig. 4(B)). It was therefore not surprising that when we measured potentiation in response to zinc or acidic pH at a fixed concentration of ATP (Fig. 4(C)), the potentiation ratios were greater at low expression levels. In these experiments we used 10 μ M

ATP, to facilitate comparison with the data shown for wild type $P2X_2$. Under these conditions, the maximum potentiation of H397# responses (up to 7-fold) was less than the maximum potentiation of wild type $P2X_2$ (up to 20-fold). The explanation for this difference is that although the range of EC_{50} values for wild type $P2X_2$ and H397# overlapped considerably, the average concentration–response relation of H397# was slightly left-shifted with respect to wild type $P2X_2$ (mean EC_{50} of 22 μ M for wild type and 14 μ M for H397#). Therefore, on average, 10 μ M ATP is a higher effective concentration for H397# than for the wild type, and so less potentiation is possible.

4. Discussion

We found that the EC_{50} of the ATP concentration–response relation of rat $P2X_2$ receptors was inversely related to the level of receptor expression, but that the Hill coefficient did not change with expression level. Because zinc and pH potentiation result from a shift in the concentration–response relation with little or no change in the maximum current (King et al., 1996, 1997; Nakazawa et al., 1997; Stoop et al., 1997; Clyne et al., 2002), the amount of zinc and pH potentiation also depended on expression levels. For example, if the response of $P2X_2$ receptors to 10 μ M ATP corresponds to the EC_{10} in an oocyte with low receptor expression but corresponds to the EC_{40} in another oocyte with high expression, then the most potentiation that can be achieved in these two oocytes is 10- and 2.5-fold, respectively.

The inverse correlation between potentiation ratios and expression levels was a strong trend, but there were exceptions. In particular, some low expressing oocytes demonstrated relatively little potentiation. We demonstrated in Fig. 3G that the potentiation ratio was extremely sensitive to small deviations from the EC_{10} , and so there is unavoidable variation over the range of about 5- to 12-fold in the measured potentiation ratios when one intends to use the EC_{10} concentration. However, this variation is uncorrelated with expression level. In contrast, the dependence of EC_{50} on expression level imparts a strong trend to the potentiation data, yielding potentiation ratios to 10 μ M ATP as large as 25-fold for some low expressing oocytes, and as small as 1.4-fold for some high expressing oocytes.

What might cause the properties of $P2X_2$ receptors to change with receptor expression levels? A simple model is that $P2X_2$ receptors can exist in either of two states with different EC_{50} values, and the proportion of receptors in these states changes with receptor expression levels. Fig. 5 presents two molecular mechanisms that might shift the proportion of the receptors in the two states with changes in expression levels.

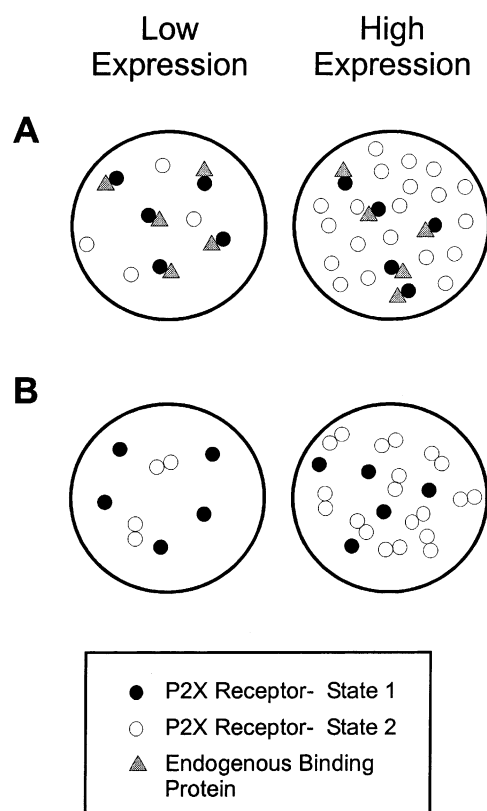


Fig. 5. Two possible models to account for the changes in receptor properties with expression levels. Each large circle represents a small patch of membrane containing P2X₂ receptors that can exist in two states. State 1 corresponds to receptors with a larger EC₅₀ (e.g. 35 μ M), state 2 corresponds to receptors with a lower EC₅₀ value (e.g. 5 μ M). Increasing receptor expression (right panel) increases the proportion of receptors in state 2. (A) Model 1: Interactions between P2X₂ receptors and an endogenous binding protein (gray triangle) stabilizes state 1 of P2X₂ receptors (black circles). At low levels of expression, most P2X₂ receptors are associated with the binding protein (left panel). Increasing the expression of P2X₂ receptors saturates the binding protein, leaving many receptors unbound and in state 2 (open circles). (B) Model 2: The clustering of P2X₂ receptors stabilizes state 2 of P2X₂ receptors. At low expression levels (left panel) few aggregates form and so most receptors are in state 1. Increasing expression of P2X₂ receptors (right panel) promotes aggregation and the number of receptors in state 2.

One possible mechanism (Model 1) is that P2X₂ receptors might complex with one or more binding proteins that are present at a limiting concentration (Fig. 5(A)) and that bound receptors have a high EC₅₀ (state 1) and unbound receptors a low EC₅₀ (state 2). Receptors associated with binding proteins would be expected to be clustered if the binding proteins could associate with more than one receptor. Indeed, a growing body of evidence suggests that P2X receptors, like other ligand-gated receptors, are clustered on the plasma membrane of neurons and muscle cells. For example, clusters of P2X₁ receptors of about 0.4–1 μ m in diameter have been observed by immunocytochemistry on the smooth muscle cells of the vas deferens, bladder, arteries, as well as

on sensory neurons of the dorsal root ganglia and sympathetic neurons of the superior cervical ganglion (Barden et al., 1999; Dutton et al., 1999; Hansen et al., 1999; Barden and Bennett, 2000; Li et al., 2000; Pulviranti et al., 2000; Liang et al., 2001). P2X₁ receptors can also form clusters when expressed in *Xenopus* oocytes and HEK293 cells (Dutton et al., 2000). In addition, P2X₁ and P2X₃ hexamers, nonamers, and higher multiples of three were detected in *Xenopus* oocytes by blue native PAGE (Nicke et al., 1998). Clustering of P2X receptors could be mediated by adapter proteins similar to those that cluster other ligand-gated receptors such as rapsyn (Colledge and Froehner, 1998b), gephyrin (Moss and Smart, 2001), PICK1 (Xia et al., 1999), and GRIP (O'Brien et al., 1998; Colledge and Froehner, 1998a). These binding proteins might also be differentially distributed along the plasma membrane. For example, binding proteins might be localized to the animal pole of *Xenopus* oocytes, where P2X₂ receptors are preferentially targeted (Wild and Paysan, 2002). Overexpressing P2X₂ receptors might lead to spillover of the receptors toward the vegetal pole, increasing the probability of the receptor being in state 2.

Another possible mechanism (Model 2) is that P2X₂ receptors might be capable of directly interacting with each other (Fig. 5(B)). In this case, increasing expression levels would promote the formation of clusters (rather than an increase in unclustered receptors as in Model 1), and clustered receptors would represent state 2. Indeed, there is indirect evidence that P2X receptor clustering may be dependent upon receptor expression levels: P2X₁ receptors appear gradually distributed along the surface of *Xenopus* oocytes after 1 day of injection and become more aggregated over the next few days as expression levels increase (Dutton et al., 2000).

Can clustering alter the properties of other receptors? Changeux et al. (1967) proposed that ligand binding to a single receptor might induce a conformational change that spread to neighboring receptors if the receptors were organized in a two-dimensional lattice. Recently, Bray and coauthors suggested that a mixed population of clustered and isolated chemotactic membrane receptors could account for the range and sensitivity of bacterial chemotaxis (Bray et al., 1998, 2002; Duke and Bray, 1999). In their model, chemotactic receptors were hypothesized to form clusters and clustering was predicted to increase sensitivity to ligand as a function of the number of clustered receptors. This model is similar to our finding that P2X₂ receptors have a lower EC₅₀ when highly expressed.

Although our models were formulated so that the law of mass action would cause high-level expression to increase the proportion of receptors in state 2, many other factors might also contribute to the balance of the equilibrium between states 1 and 2. It is easy to imagine that reversible modifications such as protein phosphoryl-

ation might alter the equilibrium independent of expression levels. Furthermore, the actual rates of transition between state 1 and state 2 might be important. For instance, in Model 2, if the lifetime of the P2X₂ protein is long and the rate of dissociation from receptor–receptor dimers is very low compared to the rate of dimer formation, then at equilibrium most receptors will end up as dimers, even when the expression level is low. In this case the major difference between high expressing and low expressing cells would be the rate at which receptors enter state 2, which would be much slower at low expression levels. It therefore would be of interest to explore whether there is a reproducible shift in the EC₅₀ as a function of time after RNA injection, especially in oocytes in which the maximal response is relatively small at 48 h. Although the recordings we report here were carried out 2–5 days after injection, all oocytes were transferred to low temperature at 48 h after injection (to prevent excessive cell death in high expressing oocytes) and so we do not presently have information to address this question. Furthermore, if receptor number continues to rise as a function of time after injection, then it may be very difficult to separate effects of time alone from effects of the level of expression.

It is clear that the last 76 amino acids of the C-terminal of the P2X₂ receptor are not necessary for the expression level dependent change in EC₅₀. This leaves nearly 400 amino acids of the receptor untested as the site of potential interactions with other P2X₂ receptors or with accessory proteins. An observation that may be suggestive of the relevant part of the molecule was made by Kim et al. (1997). They demonstrated that a peptide consisting of only the extracellular domain of P2X₂ could form stable tetramers in solution. The relevance of this work came into question when subsequent evidence (Nicke et al., 1998; Stoop et al., 1999) strongly suggested that the subunits of a single P2X₂ channel are arranged as a trimer. It seems possible that the tetramers observed by Kim et al. (1997) were the result of interactions between an extracellular domain that promotes aggregation of P2X₂ receptors.

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